

Routing Scheme Evaluation Results

Autonomous Networks Research Group (ANRG)
University of Southern California

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1 Evaluation Setup

Parameter	Value
Seeds per combo	30 (seeds 1–30); values are mean makespan
Schedulers	HEFT: calibrated to runtime routing model HEFT-1: direct-link BW; 0.001 MB/s for non-adjacent HEFT-2: widest-path BW for all pairs (fixed)
Routing schemes	W, S, SH, GS, GC, GB, GO, GSD, GSD-D
Interference model	<code>csma_bianchi</code> (802.11ax, 5 GHz, 20 MHz)
RF parameters	$P_{tx} = 20$ dBm, path-loss exponent $n = 3.0$
Grid spacing	40 m; adjacent-link PHY rate ≈ 8.6 MB/s
Compute costs	150–1000 cu (heterogeneous, $\approx 6.7\times$ range)
Data sizes	2–30 MB (heterogeneous, $15\times$ range)
Node capacities	80–300 cu/s (heterogeneous)
Total runs	3 schedulers $\times$ 9 routing $\times$ 4 experiments $\times$ 30 seeds = 3240

Table 1: Evaluation parameters.

HEFT (calibrated) passes the runtime routing model to HEFT’s pairwise rate matrix, so placement decisions are calibrated to whichever routing scheme runs at simulation time. Equals HEFT-2 when paired with `widest_path`.

HEFT-1 pins 0.001 MB/s on all non-adjacent pairs regardless of runtime routing. Strongly biases placement toward same-node or direct-neighbour assignments.

HEFT-2 always uses `WidestPathRouting` for pairwise BW estimates, regardless of the runtime routing scheme.

SH (Shortest Hop) minimises hop count, breaking ties by $\sum 1/b_\ell$. Added alongside S to test whether reducing relay count reduces `csma_bianchi` interference enough to offset lower bottleneck bandwidth.

2 Full Results Tables

Mean makespan in seconds, averaged over 30 seeds. **Bold green**: overall experiment winner. **Red**: worst entry in that column. †: best within that scheduler column.

2.1 4×4 Grid, Small DAG (8 tasks, fork-join, 12 edges)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	1814.518	18.336	1814.518
S	93.856	18.336	93.856
SH	93.856	18.336	93.856
GS	93.319	18.336	93.319
GC	93.319	18.336	93.319
GB	93.319	18.336	93.319
GO	93.319	18.336	93.319
GSD	140.483	18.336	140.483
GSD-D	91.134[†]	18.035[†]	91.134[†]
<i>Best</i>	<i>GSD-D 91.134</i>	<i>GSD-D 18.035</i>	<i>GSD-D 91.134</i>

Table 2: 4×4 small. HEFT-1 co-locates all 8 tasks on one node; no transfers occur, so routing is irrelevant (all schemes tie at 18.336 s). GSD-D gains 0.3 s via deferral logic on same-node handoffs. HEFT and HEFT-2 are identical here (both use widest-path BW when routing=W; for other schemes the 4×4 topology’s BW estimates converge). SH $\equiv$ S throughout: grid paths that minimise $\sum 1/b$ also minimise hops on this small topology.

2.2 4×4 Grid, Large DAG (30 tasks, 5-stage pipeline, 48 edges)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	2383.795	497.788	2383.795
S	1071.723	292.425	1071.723
SH	1071.723	292.425	1071.723
GS	1026.909	292.091[†]	1026.909
GC	1066.630	292.091[†]	1066.630
GB	1156.735	292.091[†]	1156.735
GO	1022.295	292.091[†]	1022.295
GSD	855.880[†]	671.422	855.880[†]
GSD-D	1037.424	322.536	1037.424
<i>Best</i>	<i>GSD 855.880</i>	<i>GS/GC/GB/GO 292.091</i>	<i>GSD 855.880</i>

Table 3: 4×4 large. HEFT and HEFT-2 are identical: with 30 tasks all paths converge regardless of routing calibration. Under HEFT-1, static greedy schemes (GS/GC/GB/GO) all tie at 292.091 s—41% vs W—by routing direct-neighbour transfers efficiently. Within the original HEFT, GSD wins at 855.880 s: calibrated placement spreads tasks and GSD’s runtime congestion avoidance pays off. SH $\equiv$ S again: min-hop and min-delay paths coincide on this compact grid.

2.3 7×7 Grid, Small DAG (8 tasks, fork-join, 12 edges)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	1926.814	24.608	1926.814
S	93.902	20.262	93.696[†]
SH	93.819[†]	20.262	94.067
GS	1358.794	20.262	1309.776
GC	1135.942	20.262	1139.435
GB	1597.079	20.262	1545.814
GO	1494.147	20.262	1542.840
GSD	1393.943	20.262	1459.882
GSD-D	1512.787	19.779[†]	1338.502
<i>Best</i>	<i>SH 93.819</i>	<i>GSD-D 19.779</i>	<i>S 93.696</i>

Table 4: 7×7 small. The only experiment where SH and S diverge under HEFT: SH (93.819 s) edges out S (93.902 s) by 0.08 s—a 0.1% difference, within noise. Under HEFT-2, S (93.696 s) wins. Under HEFT-1 all tasks are co-located; GSD-D wins at 19.779 s via deferral. Greedy static schemes under HEFT/HEFT-2 collapse to 1100–1600 s because calibrated placement spreads tasks but greedy routing cannot handle the congestion.

2.4 7×7 Grid, Large DAG (60 tasks, 6-stage pipeline, 102 edges)

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	5695.953	1318.190	5657.358
S	2487.685[†]	618.267	2497.366[†]
SH	2536.137	656.054	2530.784
GS	4295.565	660.732	4432.067
GC	4431.863	615.080	4468.524
GB	4353.115	624.117	4511.408
GO	4439.492	542.189[†]	4361.533
GSD	4192.354	1278.159	4459.398
GSD-D	4436.056	1556.583	4474.458
<i>Best</i>	<i>S 2487.685</i>	<i>GO 542.189</i>	<i>S 2497.366</i>

Table 5: 7×7 large. **S beats SH** under both HEFT (−2%) and HEFT-1 (−6%): on this topology, S’s 2-hop high-BW paths outperform SH’s 1-hop diagonals because bottleneck bandwidth dominates over relay interference. Under HEFT-1, GO wins at 542.189 s: overlap-based ordering best handles the dense parallel transfers in the 60-task pipeline, routing the most contested flows first. GSD and GSD-D are worse than W under HEFT-1 because compact placement leaves no beneficial alternate paths. Original HEFT calibrated to S delivers 2487.685 s—5.4× slower than HEFT-1+GO.

3 Cross-Experiment Comparison

3.1 Overall Winners per Experiment

Experiment	HEFT best	(s)	HEFT-1 best	(s)	HEFT-2 best	(s)
4×4 small	GSD-D	91.134	GSD-D	18.035	GSD-D	91.134
4×4 large	GSD	855.880	GS-GO	292.091	GSD	855.880
7×7 small	SH	93.819	GSD-D	19.779	S	93.696
7×7 large	S	2487.685	GO	542.189	S	2497.366

Table 6: Best routing scheme per scheduler. HEFT-1 wins every experiment outright. HEFT and HEFT-2 are identical on 4×4 (BW estimates converge on the compact grid).

3.2 HEFT-1: Routing Rankings Across Experiments

Routing	4×4 S (s)	4×4 L (s)	7×7 S (s)	7×7 L (s)	Wins
W	18.336	497.788	24.608	1318.190	0
S	18.336	292.425	20.262	618.267	0
SH	18.336	292.425	20.262	656.054	0
GS	18.336	292.091	20.262	660.732	1
GC	18.336	292.091	20.262	615.080	1
GB	18.336	292.091	20.262	624.117	1
GO	18.336	292.091	20.262	542.189	2
GSD	18.336	671.422	20.262	1278.159	0
GSD-D	18.035	322.536	19.779	1556.583	2

Table 7: HEFT-1 column winners. GO leads with 2 wins (the highest-value experiments). GSD-D wins 2 small-DAG experiments. SH never beats S under HEFT-1; S is consistently equal or better. GSD/GSD-D are worst on large grids: compact placement leaves no beneficial alternate paths for dynamic routing.

3.3 HEFT: Routing Rankings Across Experiments

Routing	4×4 S (s)	4×4 L (s)	7×7 S (s)	7×7 L (s)	Wins
W	1814.518	2383.795	1926.814	5695.953	0
S	93.856	1071.723	93.902	2487.685	1
SH	93.856	1071.723	93.819	2536.137	1
GS	93.319	1026.909	1358.794	4295.565	0
GC	93.319	1066.630	1135.942	4431.863	0
GB	93.319	1156.735	1597.079	4353.115	0
GO	93.319	1022.295	1494.147	4439.492	1
GSD	91.134	855.880	1393.943	4192.354	2
GSD-D	91.134	1037.424	1512.787	4436.056	1

Table 8: Original HEFT (calibrated) column winners. GSD leads (2 wins), benefiting from calibrated placement that spreads tasks across the grid and runtime congestion avoidance. SH edges S by 0.08 s on 7×7 small—within noise. S wins the 7×7 large experiment by a clear margin over SH (-2%). W is always worst.

3.4 SH vs S: Does Hop-Count Routing Help?

Experiment	S (s)	SH (s)	SH vs S (%)
<i>Original HEFT (calibrated)</i>			
4 × 4 small	93.856	93.856	0.0
4 × 4 large	1071.723	1071.723	0.0
7 × 7 small	93.902	93.819	−0.1
7 × 7 large	2487.685	2536.137	+2.0
<i>HEFT-1 (penalty)</i>			
4 × 4 small	18.336	18.336	0.0
4 × 4 large	292.425	292.425	0.0
7 × 7 small	20.262	20.262	0.0
7 × 7 large	618.267	656.054	+6.1
<i>HEFT-2 (widest-path calibrated)</i>			
4 × 4 small	93.856	93.856	0.0
4 × 4 large	1071.723	1071.723	0.0
7 × 7 small	93.696	94.067	+0.4
7 × 7 large	2497.366	2530.784	+1.3

Table 9: SH vs S makespan comparison. **Bold**: lower of the two. SH never wins by a meaningful margin: the sole SH lead (7×7 small, HEFT) is 0.1%, within 30-seed averaging noise. S outperforms SH by 6% on 7 × 7 large under HEFT-1. Conclusion: **use S, not SH**, on these 40 m grids.

4 Analysis

4.1 Why HEFT-1 Dominates on csma_bianchi Grids

HEFT-1’s 0.001 MB/s penalty forces all communicating tasks onto the same node or direct neighbours. On a csma_bianchi grid, PHY rates drop steeply with distance and relay hops accumulate interference; co-location avoids both. HEFT and HEFT-2 spread tasks across the topology, where runtime interference is typically 5–10× worse than HEFT’s estimates.

4.2 Why SH Does Not Beat S on These Grids

The hypothesis was: fewer relay hops $\Rightarrow$ fewer interfering transmitters $\Rightarrow$ lower total interference $\Rightarrow$ lower makespan. The empirical result contradicts this on the 40 m grids. The reason: on a grid with 8.6 MB/s cardinal links and lower-BW diagonal links at 56.6 m, the path that minimises $\sum 1/b$ (S) typically uses 2-hop cardinal routes rather than 1-hop diagonal routes. The 2-hop path has higher bottleneck bandwidth (8.6 vs lower diagonal BW), which more than compensates for the second active link. SH’s 1-hop preference selects the diagonal when S would use a 2-hop cardinal path—SH gets fewer hops but lower throughput.

When SH might win: topologies where all alternate paths have the same bandwidth (equal-BW grids) but different hop counts, making the relay interference cost the binding constraint. On the heterogeneous-BW grids tested here, this condition is not met.

4.3 Original HEFT vs HEFT-2: When Does Calibration Matter?

On the 4×4 grid, original HEFT and HEFT-2 produce identical results for all routing schemes. The topology is compact enough that different routing models return similar path bandwidths, so the pairwise rate matrices converge.

On the 7×7 grid with the large DAG, calibration starts to matter: HEFT (calibrated to GSD) gives 4192 s for GSD-routing, while HEFT-2 gives 4460 s. The calibrated scheduler places tasks at nodes where GSD-style congestion avoidance is effective; HEFT-2’s widest-path calibration places them differently, and GSD cannot recover.

For S-routing on 7×7 large, both perform similarly (2488 vs 2497 s) because S’s path bandwidths are close to widest-path bandwidths for most pairs.

4.4 GO Wins on 7×7 Large (HEFT-1)

Under HEFT-1, tasks are placed on direct neighbours. The 60-task pipeline creates many simultaneous transfers at each stage transition. GO (overlap-based ordering) routes the most contested flows first—flows whose time windows overlap with the most other flows get first pick of low-interference paths. On the denser 7×7 grid (240 links, more routing diversity), this ordering extracts more benefit than start-time ordering (GS), delivering 542 s vs 661 s (−18%).

4.5 Key Takeaways

1. **HEFT-1 dominates on csma_bianchi grids.** Co-location is optimal; HEFT and HEFT-2 spread tasks into heavy interference.
2. **Use GO with HEFT-1 on large grids.** Overlap-based ordering outperforms start-time (GS) and byte-size (GB) by routing the most contested flows first. On small grids, GSD-D is slightly better.
3. **SH does not improve over S** on heterogeneous-BW grids. Min-delay (S) beats min-hop (SH) because 2-hop high-BW paths deliver higher bottleneck bandwidth than 1-hop low-BW diagonals.
4. **W is always the worst routing scheme.** Widest-path routes flows through the highest-BW but also most-congested paths, maximising interference under all three schedulers.
5. **Original HEFT calibration matters on large topologies.** On 7×7 , calibrated HEFT outperforms HEFT-2 for dynamic routing schemes (GSD) by correctly anticipating per-scheme path bandwidths.

5 Reproducing These Results

```
cd ncsim/  
python run_routing_eval.py  
# Output JSON: /tmp/ncsim_full_eval/eval_results.json  
# Runtime: ~590 s (3240 runs, 8 parallel workers)
```